

Overview

The model represents a population of individual agents connected by a social network. Each agent has:

- a memory of accumulated hazard impacts;
- a level of knowledge of a mitigation practice;
- a probability of using that practice.

The analytical model removes the random draws from the agent-based simulation while preserving its main mechanisms:

1. exponentially increasing hazard frequency;
2. decaying memory of hazard impacts;
3. learning from practicing neighbors;
4. forgetting when neighbor exposure is absent;
5. reduced practice use caused by practice cost.

The output is an expected trajectory for the population. It is not intended to reproduce one stochastic simulation path exactly. A stochastic simulation fluctuates around the analytical trajectory because it samples event counts, event severities, learning encounters, and practice-use decisions.

Interpretation and limitations

What the model represents well

- The direction and timing of hazard-frequency growth;
- accumulation and decay of remembered impacts;
- the effect of network connectivity on diffusion;
- the effect of repeated exposure requirements;
- gradual forgetting without social reinforcement;
- suppression of practice use due to cost.

What the current analytical model simplifies

- It uses the expected absolute Gumbel severity rather than sampling events;
- all agents receive the same expected hazard impact at each time step;
- neighbor-use events are treated as conditionally independent

The analytical model is a deterministic network approximation. A fully homogeneous mean-field model could replace the individual network states with population averages. A more detailed continuum formulation could use age- or knowledge-structured partial differential equations, especially if the distribution of learning progress is important.

Parameters

Parameter	Fidutial	Meaning
N_AGENTS N	250	Number of individual agents.
DURATION_MONTHS T	240	Number of monthly time steps.

Parameter	Fidutial	Meaning
TIME_STEP_MONTHS t_s	1	Nominal time-step size. The current script advances one month per loop.
NETWORK_TYPE n_t	"small_world"	Network type: <code>small_world</code> , <code>erdos_renyi</code> , or <code>complete</code> .
NETWORK_NEIGHBORS $n^{\mathcal{N}}$	24	Initial ring degree for the small-world network. Must be even and below <code>N_AGENTS</code> .
NETWORK_REWIRE_PROBABILITY n_r	0.05	Probability of rewiring a small-world edge.
NETWORK_EDGE_PROBABILITY n_e	0.04	Edge probability for an Erdős-Rényi network.
HAZARD_INITIAL_FREQUENCY h_0	0.01	Initial hazard-event frequency per agent per month.
HAZARD_RATE h_r	0.012	Exponential hazard-growth rate per month.
NEV_MU μ	0.0	Location parameter of the Gumbel severity distribution.
NEV_BETA β	0.1	Scale parameter of the Gumbel severity distribution.
SEVERITY_DECAY d_s	0.015	Monthly decay rate of severity memory.
KNOWN_FRACTION f_{known}	0.05	Fraction of agents that are always fully knowledgeable.
LEARNING_TIMES L_t	4.0	Learning exposures required to reach full knowledge.
PRACTICE_DECAY d_p	0.04	Monthly forgetting rate when there is no neighbor exposure.
PRACTICE_COST C	0.35	Cost of practice use.
COST_SENSITIVITY C_s	0.5	Scale controlling how strongly cost suppresses practice use.
MEMORY_RESPONSE_SCALE M_0	1.0	Scale of the response to remembered hazard impact.

Network

The population is represented by an undirected graph:

$$G = (V, E),$$

where:

- $V = \{1, \dots, N\}$ is the set of agents;
- E is the set of social connections

The analytical model retains individual values on the graph (E_i as the neighbor set of agent i .), so it is a network mean-field approximation rather than a fully homogeneous population model.

Small-world network

The default network starts as a ring lattice. Each agent is connected to $n^{\mathcal{N}}$ nearby agents, then edges are rewired with probability n_r at each time step.

Other network architectures

The implementation also supports:

- Erdős-Rényi: each possible edge is included independently with probability `NETWORK_EDGE_PROBABILITY`;
- complete: every agent is connected to every other agent.

Time and state variables

Time is measured in months:

$$t = 0, 1, 2, \dots, T,$$

where T is `DURATION_MONTHS`.

For every agent i , the analytical model computes each month:

- $S_i(t)$: remembered severity impact;
- $L_i(t)$: learning progress;
- $K_i(t)$: normalized practice knowledge;
- $U_i(t)$: expected probability of using the practice;
- $P_i(t)$: probability of exposure to at least one practicing neighbor.

The implementation stores the vectors $S(t)$, $L(t)$, and $U(t)$.

Step 1: Hazard frequency

The hazard frequency per agent grows exponentially with time:

$$h(t) = h_0 e^{h_r t},$$

where:

- h_0 is `HAZARD_INITIAL_FREQUENCY`;
- h_r is `HAZARD_RATE`;
- t is time in months.

The hazard function will be modified to be just linear growth, flat and decreasing.

The expected total number of events in the population is:

$$\mathbb{E}[N_{\text{events}}(t)] = Nh(t),$$

where N is `N_AGENTS`.

Step 2: Expected event severity

In the stochastic model, an event severity is sampled as the absolute value of a Gumbel random variable:

$$X \sim \text{Gumbel}(\mu, \beta),$$

$$S = |X|.$$

The Gumbel density is:

$$p_X(x) = \frac{1}{\beta} \exp \left[- \left(z + e^{-z} \right) \right], \quad z = \frac{x - \mu}{\beta}.$$

The analytical model computes the expected absolute severity numerically:

$$\bar{S} = \mathbb{E}[|X|] = \int_{-\infty}^{\infty} |x| p_X(x) dx.$$

on the interval:

$$[\mu - 12\beta, \mu + 12\beta].$$

Step 3: Expected monthly impact

The expected number of events per agent is $f(t)$, and the expected severity per event is $\bar{S}$. Therefore, the expected monthly impact per agent is:

$$I(t) = h(t)\bar{S}.$$

This value is deterministic and identical for all agents during a given month. The model does not sample individual event counts or severities.

Step 4: Severity-memory update

Memory decays exponentially between time steps.

The memory recurrence is:

$$S_i(t+1) = d_s S_i(t) + I(t),$$

where d_s is `SEVERITY_DECAY`.

Because all agents receive the same expected impact in the analytical solution, and start with zero memory, the analytical memory values remain equal in the current implementation, so there is no need to distinguish between agents. However, we keep the vector form for now because it allows future heterogeneous exposure or network-dependent hazard forcing:

The population-level history is:

$$\overline{S}(t) = \frac{1}{N} \sum_{i=1}^N S_i(t).$$

Step 5: Probability of neighbor exposure

Let $U_j(t)$ be the expected probability that neighbor j uses the practice. Assuming neighbor-use events are conditionally independent, the probability that agent i sees at least one practicing neighbor is:

$$P_i(t) = 1 - \prod_{j \in E_i} (1 - U_j(t)).$$

We initialize $U_j(t)$ as 0, we describe later how $U_j(t)$ is computed.

Step 6: Learning and forgetting

Learning progress is measured in exposure equivalents. Full knowledge requires **LEARNING_TIMES** L_t exposures

For an exposure event, progress advances by one exposure until L_t is reached

$$L_i^+(t) = \min(L_t, L_i(t) + 1).$$

When an agent is not exposed to a practicing neighbor, progress decays exponentially.

The no-exposure progress is:

$$L_i^-(t) = d_p L_i(t),$$

where d_p is **PRACTICE_DECAY**.

Using the previous exposure probability $P_i(t)$, the progress in learning is shaped by a mixture of exposure and forgetting:

$$L_i(t+1) = P_i(t) L_i^+(t) + (1 - P_i(t)) L_i^-(t).$$

An agent is counted as fully knowledgeable while:

$$L_i(t) \geq L_t.$$

Thus, the learned-agent fraction is:

$$F_{\text{known}}(t) = \frac{1}{N} \sum_{i=1}^N \mathbf{1}[L_i(t) \geq L_t].$$

Note that agents keep forgetting even if they have learned beyond a threshold.

Step 7: Knowledgeable, Hazard response, and Practice cost

Knowledge

An agent knowledge level is:

$$K_i(t) = 1 \text{ if } \frac{L_i}{L_t} \geq \frac{L_t - 1}{L_t},$$

$$K_i(t) = 0 \text{ otherwise.}$$

This is meant to be a threshold of knowledge, once an agent is one exposure away to have learned the practice, it is assumed to be able to learn it while practicing one more time.

The population normalized mean knowledge (between 0 and 1) is:

$$\overline{K}(t) = \frac{1}{N} \sum_{i=1}^N K(t)_i.$$

There is a step-like change in the knowledge fraction, think of a way to plot that transition time depending on the parameters: d_p , L_t , n^N , kind of network

Response

The response to remembered hazard impact is a saturating function:

$$R_i(t) = \left[1 - \exp \left(-\frac{S_i(t)}{M_0(1 - f_u)} \right) \right],$$

where M_0 is **MEMORY_RESPONSE_SCALE** and f_u is the fraction of agents using the practice.

We assume this saturating function for a stronger response when the perceived severity is big and the fraction of agents using the practice is almost total.

Cost

The affordability of the practice is:

$$A_p = \exp \left(-\frac{C}{C_s} \right),$$

where:

- C is **PRACTICE_COST**;
- C_s is **COST_SENSITIVITY**.

We selected a exponential adjustment to make A_p dimensionless, and the cost of the practice to be determined relative to a factor of susceptibility to be used. More affordable practices are more likely to be used.

Step 8: Use practice probability

The expected practice-use probability is:

$$U_i(t) = \text{clip} (K_i(t) \cdot R_i(t) \cdot A_p, 0, 1),$$

equivalently:

$$U_i(t) = \text{clip} \left[K_i(t) \left(1 - e^{-S_i(t)/(M_0 - M_0 f_u)} \right) e^{-C/C_s}, 0, 1 \right],$$

where *clip* means that values higher than 1 become 1, and lower than 0 become 0.

The analytical population-level practice-use fraction is the mean expected probability:

$$\bar{U}(t) = \frac{1}{N} \sum_{i=1}^N U_i(t).$$

Step 0: Initialization

At initialization, for a selected agent count, and network configuration, the analytical model (and stochastic model) select the same initial network and initial knowledgeable agents.

$$M_i(0) = 0, \quad L_i(0) = 0$$

For all simulations we have a fraction of agents that always know the practice, no matter what:

$$N_{\text{known}} = \text{round}(N \cdot f_{\text{known}}),$$

where f_{known} is **KNOWN_FRACTION**. We have always a minimum amount of agents that know, and could use the practice, so other agents can learn. That would be equivalent to a fraction of agents self teaching the practice or learning it from outside the simulated group.

The selected agents are assigned:

$$K_i(t) = 1.$$

The selection is random.

Step-by-step algorithm

Analytical hazard-learning dynamics

Deterministic network mean-field approximation • monthly time steps

1. Initialize model

Read N, duration, network type, seed, and parameter overrides

Build social graph $G = (V, E)$; set $S_i = L_i = U_i = 0$

Analytical approximation

Random event counts and decisions are replaced by expected values.

Individual agent states remain network indexed.

2. Compute expected event severity

$$X \sim \text{Gumbel}(\mu, \beta), S = |X|$$

$$\bar{S} = \int |x| p_X(x) dx$$

Numerical trapezoidal integration over $\mu \pm 12\beta$

3. Start month t

For $t = 0, 1, \dots, \text{DURATION_MONTHS}$

All updates below occur once per month

4. Increase hazard forcing

$$h(t) = f_0 \exp(\text{hazardRate} \cdot t)$$

$$I(t) = h(t) \cdot \bar{S} \quad (\text{expected impact per agent})$$

5. Update severity memory

$$S_i(t+1) = d_s S_i(t) + I(t)$$

6. Calculate social exposure

$$P_i(t) = 1 - \prod_{j \in N_i} (1 - U_j(t))$$

Probability that agent i sees at least one practicing neighbor

$P_i = 0$ when N_i is empty

7. Learn or forget the practice

$$L_i^+ = \min(\text{LEARNING_TIMES}, L_i + 1)$$

$$L_i^- = \exp(-\text{practiceDecay}) \cdot L_i$$

$$L_i(t+1) = P_i L_i^+ + (1 - P_i) L_i^-$$

$U(t)$ drives next month's neighbor exposure

8. Determine expected practice use

$$K_i = \min(1, L_i / \text{LEARNING_TIMES})$$

$$R_i = 1 - \exp[-M_i / (M_0 \cdot (1 - \bar{U}_{\text{previous}}))]$$

$$U_i = \text{clip}[1(K_i > (L-1)/L) \cdot R_i \cdot \exp(-\text{cost} / \text{sensitivity}), 0, 1]$$

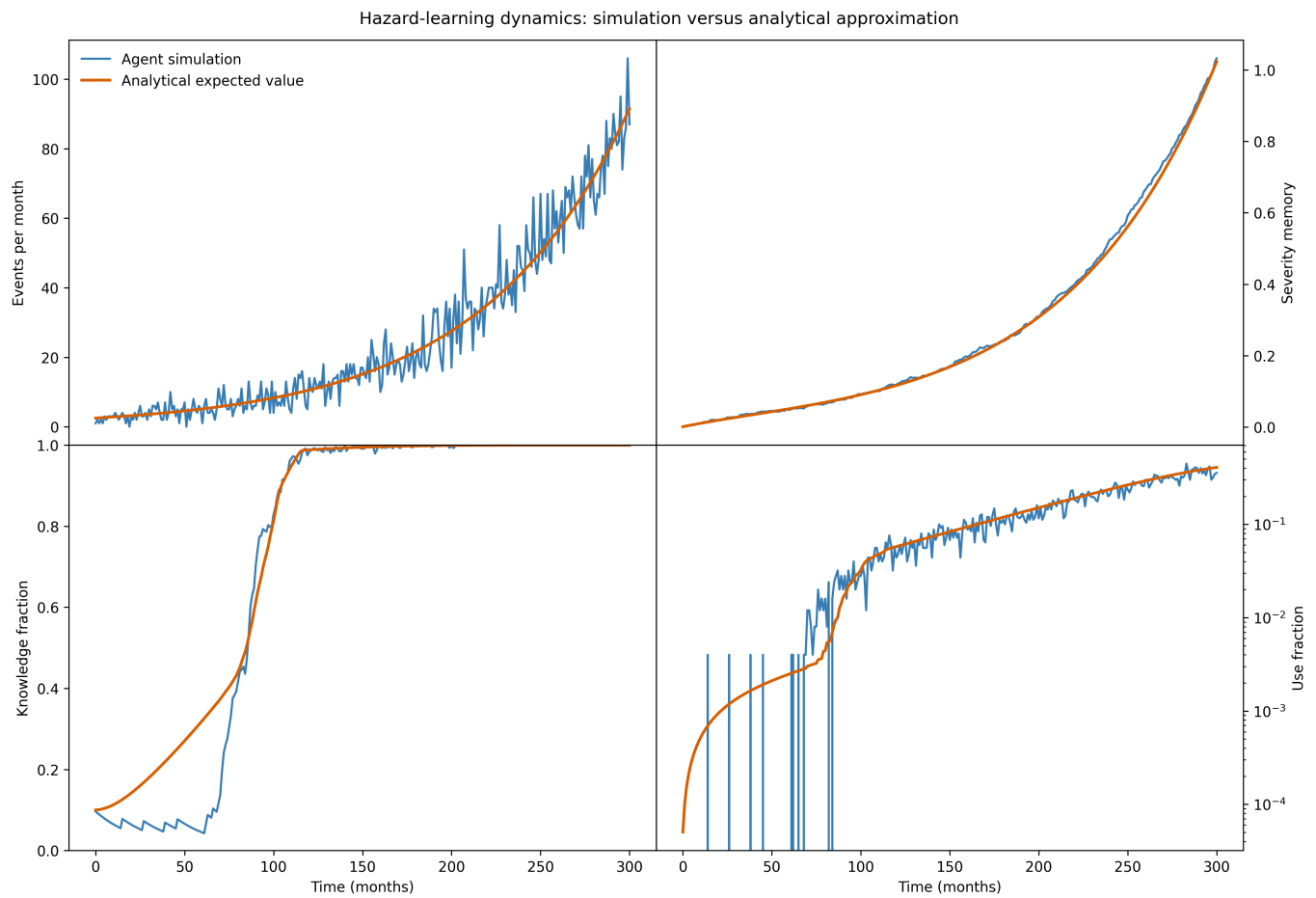
The analytical model stores probabilities, not Boolean decisions

next
month

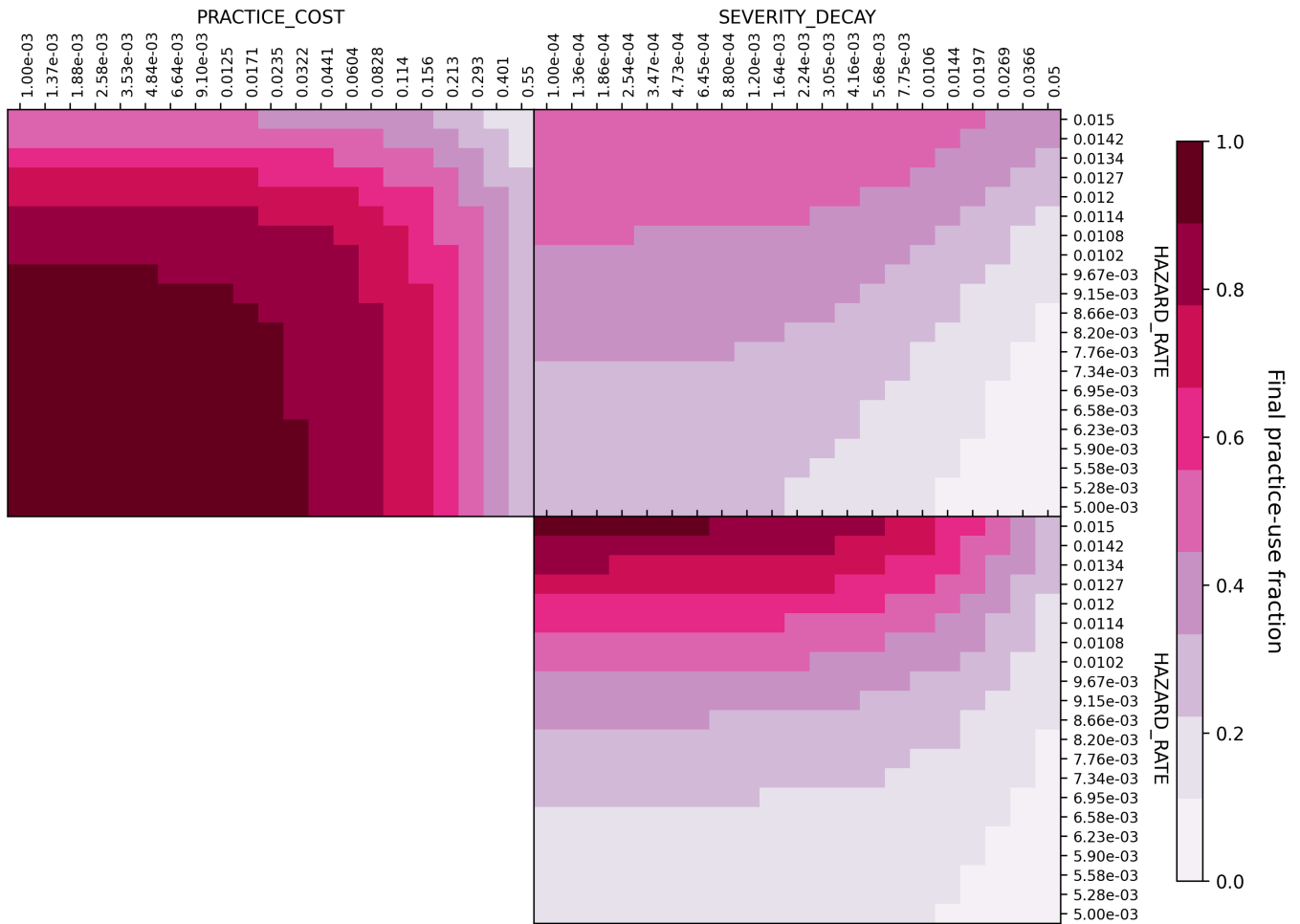
9. Record monthly histories

frequency, expected events, mean memory, mean knowledge
known fraction, practice-use fraction

Results



Results from an analytical and stochastic simulations. Top-left panel show the number of events per month. Top-right, the average remembered severity of the events. Lower-left, the fraction of the population that knows about a practice to manage the hazard. Lower-right, fraction of agents that use the practice at each month.



Use fraction of the practice after 300 months for 3 different pairs of parameters. Each panel shows -in log-log scale--the Parameter exploration for three of the main variables affecting the adoption dynamics of the practice --hazard rate, cost of the practice, memory decay of the severity.